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Estimating marginal likelihoods for mixture and Markov switching models using bridge sampling techniques*

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Summary This paper discusses the problem of estimating marginal likelihoods for mixture and Markov switching model. Estimation is based on the method of bridge sampling (Meng and Wong 1996; *Statistica Sinica 11*, 552–86.) where Markov Chain Monte Carlo (MCMC) draws from the posterior density are combined with an i.i.d. sample from an importance density. The importance density is constructed in an unsupervised manner from the MCMC draws using a mixture of complete data posteriors. Whereas the importance sampling estimator as well as the reciprocal importance sampling estimator are sensitive to the tail behaviour of the importance density, we demonstrate that the bridge sampling estimator is far more robust. Our case studies range from computing marginal likelihoods for a mixture of multivariate normal distributions, testing for the inhomogeneity of a discrete time Poisson process, to testing for the presence of Markov switching and order selection in the MSAR model.

Keywords: *Bayesian model choice*, *Bridge sampling technique*, *Marginal likelihoods*, *Markov switching models switching models*, *Mixture models*.

1. INTRODUCTION

During the past decade, one could observe an enormously increased interest in analyzing econometric data by means of nonlinear models involving discrete-valued latent variables. One prominent example is mixture models, where these latent variables are independent indicators taking K different states. Complex mixture models, such as mixtures of random-effects models, found interesting applications especially in marketing to capture unobserved heterogeneity (see e.g. Allenby *et al*. 1998; Lenk and DeSarbo 2000). Another extremely popular example are Markov switching models, where the latent variable is a hidden Markov chain with K states. Markov switching models became very popular in business cycle analysis, see, e.g. Hamilton (1989) and McCulloch and Tsay (1994), as they are rather flexible, nonlinear time series models that are able to capture asymmetric patterns found for instance by Falk (1986) in many economic time series such as GDP, investment and industrial production. Another area of application is financial econometrics, where Markov-switching models are used to account for specific features

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of financial time series such as asymmetries, fat tails and volatility clusters (see, e.g. Rydén *et al*. 1998). We mention here in particular applications to analyzing interest rates (Gray 1996) or exchange rates (Engel and Kim 1999). Hamilton and Raj (2002) provides a recent review of further applications in business cycle research and financial analysis.

Given this vast interest in applied econometric inference based on mixture and Markov switching models, it is not surprising that parameter estimation for these models is well-developed by now, either using classical methods such as the EM-algorithm, see McLachlan and Peel (2001) for a comprehensive survey of relevant references, or using Bayesian estimation via Markov chain Monte Carlo methods (MCMC), see for instance Albert and Chib (1993), Diebolt and Robert (1994), McCulloch and Tsay (1994), Shephard (1994), Chib (1996), Robert and Titterington (1998), Robert *et al*. (2000) and Frühwirth-Schnatter (2001b), for specific mixture and Markov switching models, and Chib (2001) for a general review of MCMC methods in econometrics.

One important aspect of econometric modelling, however that is not that well-understood for mixture and Markov switching models, concerns model selection problems. Following a long tradition in Bayesian econometrics initiated by Zellner (1971) in the context of selecting regression models, the marginal likelihood will be used in this paper to select among various models which differ in the number of states of the hidden latent variable, but also in other aspects of the model structure such as the model order for Markov switching autoregressive models. Marginal likelihoods have been applied to various complex econometric model selection problems, for instance by Frühwirth-Schnatter (1995), Shively and Kohn (1997) and Koop and van Dijk (2000) in the context of state space models, by Chib (1998) for change-point problems and by Kim *et al*. (1998) and Chib *et al*. (2002) to select among stochastic volatility models, see also Geweke (1995) for a general discussion of Bayesian comparison of econometric models based on marginal likelihoods. Alternative Bayesian approaches toward selecting the number of states of the latent discrete variable which are not considered here are the entropy approach of Mengersen and Robert (1996) and reversible jump Markov Chain Monte Carlo methods as used by Richardson and Green (1997) and Robert *et al*. (2000).

The computation of the marginal likelihood is in general a non-trivial integration problem (see, for instance Geweke 1999). Marginal likelihoods have been estimated using methods such as the candidate's formula (Chib 1995; Chib and Jeliazkov 2001), importance sampling based on mixture approximations (Frühwirth-Schnatter 1995), combining MCMC simulations and asymptotic approximation (Gelfand and Dey 1994; DiCiccio *et al*. 1997) and bridge sampling (Meng and Wong 1996; Meng and Schilling 2002), see the review in Chib (2001) for more detail. Although these methods proved to be useful for a wide range of econometric models, they are apt to fail when estimating the marginal likelihood of mixture or Markov switching models, as the posterior density of such a model is highly irregular due to the lack of identification for these models, see Celeux *et al*. (2000), Stephens (2000) and Frühwirth-Schnatter (2001b) for more details. It will be demonstrated in Sections 4.2 and 5.1 that in particular the estimator of the marginal likelihood suggested by Chib (1995) is sensitive to this problem, as already noted by Neal (1998).

In the present paper we investigate the technique of bridge sampling (Meng and Wong 1996) as an alternative method of computing the marginal likelihood of mixture and Markov switching models. Bridge sampling generalizes the method of importance sampling which has been applied to various complex econometric inference problems by, among others, Kloek and van Dijk (1978), van Dijk and Kloek (1980), Geweke (1989), see also Mariano *et al*. (2000). Like importance sampling, bridge sampling is based on an i.i.d. sample from an importance density, however, this sample is combined with the MCMC draws from the posterior density in an appropriate way. One

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might wonder, why this extension is sensible. For importance sampling it is well-known that the variance of the resulting estimator depends on the ratio of the non-normalized posterior density over the importance density which may be unbounded for poorly chosen importance densities (see, e.g. Geweke 1989; Geweke 1999). An important advantage of bridge sampling is, that the variance the resulting estimator depends on a ratio that is bounded regardless of the tail behaviour of the underlying importance density. This allows the econometrician far more flexibility in the construction of the importance density. In the present paper, for instance, the importance density is construct during MCMC sampling in an unsupervised manner as a mixture of complete data posteriors in a similar way as was done in Frühwirth-Schnatter (1995) for state space models. The empirical case studies will demonstrate that the bridge sampling technique is particularly useful for mixture and Markov switching models.

The outline of the rest of the paper is as follows. Section 2 reviews bridge sampling techniques for estimating marginal likelihoods, whereas Section 3 deals with the application of bridge-sampling technique to mixture and Markov switching model. Section 4 demonstrates that alternative estimators of the marginal likelihood like Chib's method (Chib 1995) and estimating marginal likelihoods for constrained models as in Lenk and DeSarbo (2000) may be problematic. Section 5 contains two empirical case studies, one of them being testing for the inhomogeneity of a discrete time Poisson process, and the second being testing for the presence of Markov switching and order selection in the Markov switching autoregressive model. Section 6 concludes the paper with some final remarks.

2. BRIDGE SAMPLING TECHNIQUES FOR ESTIMATING MARGINAL LIKELIHOODS

Bridge sampling was introduced into statistics by Meng and Wong (1996) as a simulation based technique for computing ratios of normalizing constants. DiCiccio *et al.* (1997) suggested, for rather simple model-selection problems, applying bridge sampling to the problem of computing marginal likelihoods. This section provides a review of bridge sampling techniques for estimating a marginal likelihood defined by

$$p(y) = \int p(y_1, \ldots, y_N|\phi)p(\phi)\nu(d\phi), \tag{1}$$

where $\phi \in \Phi$ is some unknown parameter vector taking values in the parameter space Φ, and ν is an appropriate measure on a sigma algebra on Φ. In what follows it is assumed that $p(\phi)$ is a proper prior, and that an explicit expression for the likelihood $p(y|\phi) = p(y_1, \ldots, y_N|\phi)$ is available, where $y = (y_1, \ldots, y_N)$ are the dependent observations.

2.1. The method

It is obvious from (1) that the marginal likelihood is equal to the normalizing constant of the posterior density:

$$p(\phi|y) = \frac{p^\star(\phi|y)}{p(y)}, \tag{2}$$

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where $p^\star(\phi|y)$ is the non-normalized posterior:

$$p^\star(\phi|y) = p(y|\phi)p(\phi).$$

Let $q(\phi)$ be a probability density function with known normalizing constant, which is some simple approximation to the posterior $p(\phi|y)$. $q(\phi)$ is called importance density for reasons that will become clear later on. Let $\alpha(\phi)$ be an arbitrary function such that

$$C^\alpha = \int \alpha(\phi)p(\phi|y)q(\phi)\nu(d\phi) > 0. \tag{3}$$

Bridge sampling is based on the following result:

$$1 = \frac{\int \alpha(\phi)p(\phi|y)q(\phi)\nu(d\phi)}{\int \alpha(\phi)q(\phi)p(\phi|y)\nu(d\phi)} = \frac{E_q(\alpha(\phi)p(\phi|y))}{E_p(\alpha(\phi)q(\phi))},$$

where $E_g(f(\phi))$ is the expectation of $f(\phi)$ with respect to the density $g(\phi)$. Substituting $p(\phi|y)$ by (2) yields the key identity for bridge sampling:

$$p(y) = \frac{E_q(\alpha(\phi)p^\star(\phi|y))}{E_p(\alpha(\phi)q(\phi))}. \tag{4}$$

To estimate the marginal likelihood for a given function $\alpha(\phi)$, the expectations on the right-hand side of (4) are substituted by sample averages, using Markov chain Monte Carlo draws $\phi^{(m)}$, $m = 1, \ldots, M$ from the posterior $p(\phi|y)$ and i.i.d. draws $\tilde{\phi}^{(l)}$, $l = 1, \ldots, L$ from the importance density $q(\phi)$:

$$\hat{p}(y) = \frac{\hat{E}_q(\alpha(\phi)p^\star(\phi|y))}{\hat{E}_p(\alpha(\phi)q(\phi))} = \frac{L^{-1}\sum_{l=1}^{L}\alpha\left(\tilde{\phi}^{(l)}\right)p^\star\left(\tilde{\phi}^{(l)}|y\right)}{M^{-1}\sum_{m=1}^{M}\alpha\left(\phi^{(m)}\right)q\left(\phi^{(m)}\right)}. \tag{5}$$

The resulting estimator $\hat{p}(y)$ is called the general bridge-sampling estimator.

As pointed out by Meng and Wong (1996), various more common methods of estimating the marginal likelihood result as special cases of the general bridge sampling estimator (5) by appropriate choices of $\alpha(\phi)$. If one uses $\alpha(\phi) = 1/q(\phi)$, for instance, one obtains from

$$p(y) = E_q\left(\frac{p^\star(\phi|y)}{q(\phi)}\right)$$

the following importance sampling estimator for the marginal likelihood used, e.g. in Frühwirth-Schnatter (1995):

$$\hat{p}_{IS}(y) = \frac{1}{L}\sum_{l=1}^{L}\frac{p^\star\left(\tilde{\phi}^{(l)}|y\right)}{q\left(\tilde{\phi}^{(l)}\right)}. \tag{6}$$

Note that (6) is based only on an i.i.d. sample $\tilde{\phi}^{(1)}, \ldots, \tilde{\phi}^{(L)}$ from the importance density $q(\phi)$. If one uses $\alpha(\phi) = 1/p^\star(\phi|y)$, the basic identity for reciprocal importance sampling (Gelfand and Dey 1994) results:

$$p(y) = \left\{E_p\left(\frac{q(\phi)}{p^\star(\phi|y)}\right)\right\}^{-1},$$

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leading to the following reciprocal importance sampling estimator:

$$\hat{p}_{RI}(y) = \left\{ M^{-1} \sum_{m=1}^{M} \frac{q\left(\phi^{(m)}\right)}{p^{\star}\left(\phi^{(m)}|y\right)} \right\}^{-1}. \tag{7}$$

Note that this time the estimator (7) is based only on the MCMC draws $\phi^{(m)}$, $m = 1, \ldots, M$ from the posterior. The importance density is only evaluated, however no draws from this density are required.

Meng and Wong (1996) discuss an asymptotically optimal choice of $\alpha(\phi)$, which minimizes the expected relative error of the estimator $\hat{p}(y)$ for i.i.d. draws from $p(\phi|y)$ and $q(\phi)$:

$$\alpha(\phi) \propto \frac{1}{Lq(\phi) + Mp(\phi|y)}.$$

We will refer to the corresponding estimator as the bridge-sampling estimator $\hat{p}_{BS}(y)$ subsequently. As the choice of $\alpha(\phi)$ depends on the *normalized* posterior $p(\phi|y)$, Meng and Wong (1996) apply the following iterative procedure to obtain $\hat{p}_{BS}(y)$ as the limit of a sequence $\hat{p}_{BS,t}(y)$. Based on an estimate $\hat{p}_{BS,t-1}(y)$ of the normalizing constant, the posterior is normalized,

$$\hat{p}(\phi|y) = \frac{p^{\star}(\phi|y)}{\hat{p}_{BS,t-1}(y)},$$

and a new estimate $\hat{p}_{BS,t}(y)$ is computed by (5). This leads to the following recursion:

$$\hat{p}_{BS,t}(y) = \hat{p}_{BS,t-1}(y) \frac{L^{-1} \sum_{l=1}^{L} \frac{\hat{p}\left(\tilde{\phi}^{(l)}|y\right)}{Lq\left(\tilde{\phi}^{(l)}\right) + M\hat{p}\left(\tilde{\phi}^{(l)}|y\right)}}{M^{-1} \sum_{m=1}^{M} \frac{q\left(\phi^{(m)}\right)}{Lq\left(\phi^{(m)}\right) + M\hat{p}\left(\phi^{(m)}|y\right)}}. \tag{8}$$

Either (6) or (7) may be used as starting value $\hat{p}_{BS,0}(y)$. Iteration (8) is typically very fast in practice.

2.2. *Relative mean-squared errors*

The performance of the various estimators described in the previous section may be measured in terms of the expected relative mean-squared error (RE^2) as in Chen *et al*. (2000):

$$RE^2(\hat{p}(y)) = \frac{E(\hat{p}(y) - p(y))^2}{p(y)^2}. \tag{9}$$

In (9) the data y are considered to be fixed, whereas $\hat{p}(y)$, being the ratio of the two random variables $\hat{E}_q(\alpha(\phi)p^{\star}(\phi|y))$ and $\hat{E}_p(\alpha(\phi)q(\phi))$, is a random variable, depending on the random sequences $\tilde{\phi}^{(1)}, \ldots, \tilde{\phi}^{(L)}$ and $\phi^{(1)}, \ldots, \phi^{(M)}$. The expectation in (9) is taken with respect to the joint distribution of these random sequences. Chen *et al*. (2000) derived approximate relative mean-squared errors under the assumption that both random sequences are i.i.d. sequences, with marginal densities $q(\phi)$ and $p(\phi|y)$, respectively. In the present context only the sample from $q(\phi)$ is an i.i.d. sequence, whereas the random sequence produced by MCMC is an autocorrelated process. Subsequently the results of Chen *et al*. (2000) are extended to account explicitly for the autocorrelation in the MCMC draws. As in Chen *et al*. (2000), approximate relative mean-squared errors are derived by the δ-method.

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For the general bridge-sampling estimator $\hat{p}(y)$ defined in (5), the following holds:

$$\frac{\hat{p}(y) - p(y)}{p(y)} = \frac{\hat{E}_q(\alpha(\phi)p^\star(\phi|y))/p(y)}{\hat{E}_p(\alpha(\phi)q(\phi))} - 1 = \frac{\hat{E}_q(\alpha(\phi)p(\phi|y))}{\hat{E}_p(\alpha(\phi)q(\phi))} - 1. \tag{10}$$

The random variables $\hat{E}_q(\alpha(\phi)p(\phi|y))$ and $\hat{E}_p(\alpha(\phi)q(\phi))$ are consistent estimators of the expectations $E_q(\alpha(\phi)p(\phi|y))$ and $E_p(\alpha(\phi)q(\phi))$, which are both equal to the integral C_α defined in (3):

$$C_\alpha = E_q(\alpha(\phi)p(\phi|y)) = E_p(\alpha(\phi)q(\phi)).$$

To derive the expectation of (10), the ratio of the two random variables $\hat{E}_q(\alpha(\phi)p(\phi|y))$ and $\hat{E}_p(\alpha(\phi)q(\phi))$ is approximated by a bivariate first-order Taylor expansion around (C_α, C_α):

$$\frac{\hat{p}(y) - p(y)}{p(y)} \approx \frac{\hat{E}_q(\alpha(\phi)p(\phi|y)) - C_\alpha}{C_\alpha} - \frac{\hat{E}_p(\alpha(\phi)q(\phi)) - C_\alpha}{C_\alpha}.$$

As the contributions to the sum of the estimator $\hat{E}_q(\alpha(\phi)p(\phi|y))$ are independent, we obtain for large L:

$$E(\hat{E}_q(\alpha(\phi)p(\phi|y)) - C_\alpha)^2 \approx \frac{1}{L}V_q(\alpha(\phi)p(\phi|y)),$$

where $V_g(f(\phi))$ is the expectation of $f(\phi)$ with respect to the density $g(\phi)$. As the contributions $f^{(m)} = \alpha\ (\phi^{(m)})q(\phi^{(m)})$ to the sum of the estimator $\hat{E}_p(\alpha(\phi)q(\phi))$ are autocorrelated, we obtain for large M (see, e.g. Geweke 1992):

$$E(\hat{E}_p(\alpha(\phi)q(\phi)) - C_\alpha)^2 \approx \frac{\rho_f(0)}{M}V_p(\alpha(\phi)q(\phi)),$$

where $\rho_f(0)$ is the normalized spectral density of the process $f^{(m)}$ at the frequency 0. $\rho_f(0)$ would be 1 for an i.i.d. sequence. Therefore

$$\hat{RE}^2(\hat{p}(y)) = \frac{1}{L}\frac{V_q(\alpha(\phi)p(\phi|y))}{C_\alpha^2} + \frac{\rho_f(0)}{M}\frac{V_p(\alpha(\phi)q(\phi))}{C_\alpha^2}. \tag{11}$$

For the special choices of $\alpha(\cdot)$ discussed in the previous section, (11) simplifies to

$$\begin{aligned}
\hat{RE}^2(\hat{p}_{IS}(y)) &= \frac{1}{L}V_q\left(\frac{p(\phi|y)}{q(\phi)}\right) \\
\hat{RE}^2(\hat{p}_{RI}(y)) &= \frac{\rho_{f_1}(0)}{M}V_p\left(\frac{q(\phi)}{p(\phi|y)}\right) \\
\hat{RE}^2(\hat{p}_{BS}(y)) &= \frac{1}{L}\frac{V_q\left(\dfrac{p(\phi|y)}{w_q q(\phi) + w_p p(\phi|y)}\right)}{E_q^2\left(\dfrac{p(\phi|y)}{w_q q(\phi) + w_p p(\phi|y)}\right)} + \frac{\rho_{f_2}(0)}{M}\frac{V_p\left(\dfrac{q(\phi)}{w_q q(\phi) + w_p p(\phi|y)}\right)}{E_p^2\left(\dfrac{q(\phi)}{w_q q(\phi) + w_p p(\phi|y)}\right)},
\end{aligned} \tag{12}$$

where $w_q = L/(L+M)$, $w_p = M/(L+M)$, $f_1(\phi) = q(\phi)/p(\phi|y)$ and $f_2(\phi) = q(\phi)/(w_q q(\phi) + w_p p(\phi|y))$.

These results allow a deeper understanding of the circumstances under which the various estimators will be stable or unstable. If the tails of the importance function $q(\phi)$ are thinner than

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the tails of the posterior $p(\phi|y)$ in some direction, then in (12):

$$\frac{p}{q} \to \infty, \quad \frac{q}{p} \to 0, \quad \frac{p}{w_p p + w_q q} \to \frac{1}{w_p}, \quad \frac{q}{w_p p + w_q q} \to 0.$$

As the ratio p/q is unbounded, the importance sampling estimator is expected to be unstable and to exhibit high-standard errors compared to the two other estimators for which the relevant ratios are bounded.

If, on the other hand, the tails of the importance function $q(\phi)$ are fatter than the tails of the posterior $p(\phi|y)$ in some direction, then

$$\frac{p}{q} \to 0, \quad \frac{q}{p} \to \infty, \quad \frac{p}{w_p p + w_q q} \to 0, \quad \frac{q}{w_p p + w_q q} \to \frac{1}{w_L}.$$

As this time the ratio q/p is unbounded, the reciprocal importance sampling estimator is expected to be unstable and to exhibit high standard errors compared to the two other estimators for which the relevant ratios are bounded.

This last finding allows further comments on the harmonic mean estimator (Newton and Raftery 1994) which is commonly known to be unstable. The harmonic mean estimator is that special case of the reciprocal importance sampling estimator (7) where $q(\phi)$ is equal to the prior $p(\phi)$:

$$\hat{p}_{HM}(y) = \left\{ M^{-1} \sum_{m=1}^{M} \frac{1}{p\left(y|\phi^{(m)}\right)} \right\}^{-1}, \tag{13}$$

with the likelihood $p(y|\phi) = p(y_1, \ldots, y_N|\phi)$ being evaluated at the posterior draws of ϕ. Consequently, whenever the prior is likely to have fatter tails than the posterior, we may rule out the harmonic mean estimator (13) as a reliable device of estimating marginal likelihoods.

To sum up, both importance sampling as well as reciprocal importance sampling are sensitive to the tail behaviour of the importance density $q(\phi)$ relative to the posterior $p(\phi|y)$. The bridge sampling estimator $\hat{p}_{BS}(y)$, however, is much more robust in this respect. Even for importance functions having fat tails in one direction and thin tails in the other, the ratios appearing in (12) are bounded. This robustness of the 'optimal' bridge-sampling estimator against the tail behaviour of the importance density is crucial especially for unsupervised choices of $q(\phi)$.

2.3. Simulation standard errors

The expected relative mean-squared error (RE^2) is a useful approximation to the expected absolute mean-squared error of $\log \hat{p}(y)$ (see Chen *et al.* 2000):

$$E(\log \hat{p}(y) - \log p(y))^2 \approx RE^2(\hat{p}(y)). \tag{14}$$

In order to assess the accuracy of the various estimators of the marginal likelihood in practice, it is necessary to estimate unknown quantities appearing in (12) such as the spectral density at 0 or various variances. In our case studies we use the sample variance of $f^{(l)} = f(\tilde{\phi}^{(l)}), l = 1, \ldots, L$ to estimate any variance $V_q(f(\phi))$ with respect to q, whereas we use the sample variance of $f^{(m)} = f(\phi^{(m)}), m = 1, \ldots, M$ to estimate any variance $V_p(f(\phi))$ with respect to p. Note that $f^{(l)}, l = 1, \ldots, L$ and $f^{(m)}, m = 1, \ldots, M$ are exactly the functional evaluations available from computing

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the various estimators. To estimate $\rho_f(0)$ we use

$$\widehat{\rho_f(0)} = 1 + 2 \cdot \sum_{s=1}^{S} \left(1 - \frac{s}{q+1}\right) r_s, \qquad r_s = M^{-1} \sum_{m=s+1}^{M} \frac{\left(f^{(m)} - \overline{f}\right)\left(f^{(m-s)} - \overline{f}\right)}{s_f^2},$$

where $\overline{f}$ and s_f^2 are the sample mean and the sample variance of $f^{(m)}$, see Chib (1995) for details.

3. ESTIMATING MARGINAL LIKELIHOODS FOR MIXTURE AND MARKOV SWITCHING MODELS

3.1. Model definition

Suppose that $y = (y_1, \ldots, y_N)$ is a sequence of N univariate or multivariate observations y_t taking either discrete or continuous values. In what follows, we assume that the distribution $p(y_1, \ldots, y_N | I^N, \theta)$ of y, which is indexed by model parameter θ, depends on a hidden sequence $I^N = (I_1, \ldots, I_N)$, where I_t is a latent (unobservable) variable taking values in the discrete space $\{1, \ldots, K\}$. To complete model definition, the distribution $p(I^N|\eta)$ of the latent process I^N has to be characterized up to a hyperparameter η. Both the classical mixture model, as well as the Markov switching autoregressive model are obtained as special cases of this model.

Subsequently we will use $\psi = (I^N, \theta, \eta) \in \mathcal{A}$ to denote the augmented parameter vector summarizing all unknown quantities in the mixture or switching model including all unknown model parameters and the latent indicators I^N.

3.2. Marginal likelihoods for mixture and Markov switching models

The marginal likelihood of a mixture or Markov switching model is defined as the normalizing constant of the unconstrained posterior $p^\star(\psi|y)$:

$$p^\star(\psi|y) = p(y_1, \ldots, y_N|\psi)p(\psi), \tag{15}$$

where $p(\psi) = p(I^N|\eta)\ p(\theta, \eta)$ is the joint prior of ψ and $p(\theta, \eta)$ is a proper prior which is invariant to relabelling the states of I_t. Then

$$p(y) = \int_{\mathcal{A}} p(y_1, \ldots, y_N|\psi)p(\psi)\nu(d\psi), \tag{16}$$

with ν being an appropriate measure on a sigma algebra on $\mathcal{A}$.

One could estimate the marginal likelihood (16) by the bridge sampling technique of Section 2 directly by choosing $\phi = (I^N, \theta, \eta)$. For mixture and switching models, however, the dimensionality of (16) can often be reduced by solving the integration with respect to part of ψ analytically and using numerical methods only for the remaining parameters. Substantial reduction of dimensionality is possible by marginalizing over I^N:

$$p(y) = \int p(y_1, \ldots, y_N|\theta, \eta)p(\theta, \eta)d(\theta, \eta), \tag{17}$$

before applying the bridge sampling technique of Section 2 with $\phi = (\theta, \eta)$. Marginalizing over I^N is possible for mixture and Markov switching model, whenever the memory of y_t with respect

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to I_t is finite,

$$p\big(y_t|y^{t-1}, I^t, \theta\big) = p\big(y_t|y^{t-1}, I_t, \ldots, I_{t-s+1}, \theta\big),$$

for some s not depending on t (see, e.g. Hamilton 1994; Chib 1996). This class encompasses many important mixture and switching models such as univariate and multivariate normal mixture models, mixtures of random effects models (Lenk and DeSarbo 2000; Frühwirth-Schnatter *et al*. 2004), and the Markov switching autoregressive model (McCulloch and Tsay 1994), but also non-conjugate models such as switching ARCH-models (Hamilton and Susmel 1994; Kaufmann and Frühwirth-Schnatter 2002).

For conditionally conjugate models, where

$$\int p\big(y_1, \ldots, y_N|I^N, \theta\big)p(\theta)\, d\theta$$

may be solved analytically, one could alternatively marginalize over the parameters (θ, η) as in Chen and Liu (1996)

$$p(y) = \int p\big(y_1, \ldots, y_N|I^N\big)p(I^N)\nu(dI^N). \tag{18}$$

It is then possible to apply the bridge sampling technique of Section 2 with $\phi = I^N$ in which case the methods suggested recently in Casella *et al*. (2000) for importance sampling of I^N will turn out to be useful. We leave, however, the details of such an approach for future research, as we believe that the drastic reduction of the dimensionality in (17) compared to (18) will facilitate the estimation of the unconstrained marginal likelihood (16). Furthermore, the application the bridge sampling technique to (17) is more general, as conditional conjugacy of the observation distribution in θ is not a requirement for marginalization over I^N.

3.3. MCMC estimation

We assume that an MCMC sampler is available which produces draws $\psi^{(m)}$, $m = 1, \ldots, M$, from the unconstrained posterior $p(\psi|y)$ (see, e.g. Frühwirth-Schnatter 2001b, and the references in there for a review of MCMC sampling for mixture and switching models). It is essential to use an MCMC sampler which explores all modes of the unconstrained posterior (15) and to avoid sampler that are not well-mixing over all possible labelling subspaces. Celeux *et al*. (2000) and Frühwirth-Schnatter (2001b) demonstrate that traditional MCMC sampling methods often lack adequate mixing and do not explore the whole support of the target distribution. As will be demonstrated in Section 4.3, estimates of the marginal likelihood which are obtained from draws of a poorly mixing sampler are prone to be biased. In our case studies we are going to apply the random permutation sampler where unconstrained sampling is concluded by a random permutation of the current labelling (Frühwirth-Schnatter 2001b), however, the methods to be discussed in this section could be based on MCMC draws of any sampler that is well-mixing with respect to the unconstrained posterior, such as the sampler of Celeux *et al*. (2000).

3.4. Choice of the importance density

To apply the bridge sampling technique described in Section 2 to estimate the marginal likelihood in (17), we have to select an importance density $q(\phi) = q(\theta, \eta)$. As manual tuning of the importance

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density for each model among the whole set of possible models is rather tedious, we need a method for choosing sensible importance densities in an unsupervised manner. Various suggestions of constructing an unsupervised importance function from the MCMC output have been made earlier. DiCiccio *et al.* (1997), for instance, suggest to construct a normal importance density from the MCMC output, however, they pointed out that this will work only for well-behaved, regular problems with unimodal posterior densities. The multimodality of the unconstrained posterior of mixture or Markov switching models evidently rules this method out. Lenk and DeSarbo (2000) fit a non-normal importance density to the MCMC draw of a constrained sampler, however, as will be demonstrated in Section 4.3 this method is sensitive to a poorly chosen constraint.

In the present paper we extend the unsupervised importance density suggested in Frühwirth-Schnatter (1995) for Gaussian state space models to mixture and Markov switching models. Construction is based on the incomplete data structure of the model. Let $K(\psi|\psi')$ denote the density of the transition kernel of the MCMC sampler w.r.t. ν:

$$p(\psi|y) = \int_{\mathcal{A}} K(\psi|\psi')p(\psi'|y)\nu(d\psi').$$

For the rest of this section we will restrict the discussion to the 'Gibbsian' case, where the density $K(\psi|\psi')$ takes following form:

$$K(\psi|\psi') = K_\theta(\theta|I^N, y, \theta')p(\eta|I^N)p(I^N|(\theta, \eta)', y), \tag{19}$$

and sampling from the density $K_\theta(\theta|I^N, y, \theta')$ involves standard densities with explicit normalizing constants, only. θ need not to be simulated at once, splitting θ into D blocks $(\theta_1, \ldots, \theta_D)$ is possible here

$$K_\theta(\theta|I^N, y, \theta') = \prod_{d=1}^{D} p\left(\theta_d|\theta_{j<d}, \theta'_{j>d}, I^N, y\right). \tag{20}$$

Extensions which relax these assumptions will be discussed in Section 6.

From (20) the importance density is constructed as a mixture density derived from the MCMC draws $\psi^{(1)}, \ldots, \psi^{(S)}$ by Rao-Blackwellisation (Gelfand and Smith 1990):

$$q(\theta, \eta) = \frac{1}{S}\sum_{n=1}^{S} p\left(\eta|(I^N)^{(n)}\right)\prod_{d=1}^{D} p\left(\theta_d|\theta_{j<d}^{(n)}, \theta_{j>d}^{(n-1)}, (I^N)^{(n)}, y\right). \tag{21}$$

Construction of the mixture approximation (21) is fully automatic and may be easily incorporated into MCMC sampling (see Frühwirth-Schnatter 2001a, p.39 for further details). As the construction is based on averaging over the conditional densities, where the indicator process I^N is sampled from the unconstrained posterior with balanced label switching, the mixture importance density (21) will be multi-modal. To illustrate this we include Figure 1 which explores the bivariate marginals of an importance density that will be used in Section 5.1 for estimating the marginal likelihood for a 3-state inhomogeneous Poisson process.

In order to reproduce all modes of the posterior it is essential to base the construction of the importance density on an MCMC method which forces balanced label switching. In the case of unbalanced label switching the resulting importance density would vanish on non-visited modal regions. An informal test, if all modes were visited, is the following: consider all permutation $\rho_r(\theta^\star, \eta^\star)$ of the components the posterior mode estimator $(\theta^\star, \eta^\star)$ and compute the ratio $q(\rho_r(\theta^\star, \eta^\star))/q(\theta^\star, \eta^\star)$ for all possible permutations of the labelling. Any ratio close to 0 indicates a non-visited mode, whereas ratios close to 1 indicate balanced switching to this mode. Based on these

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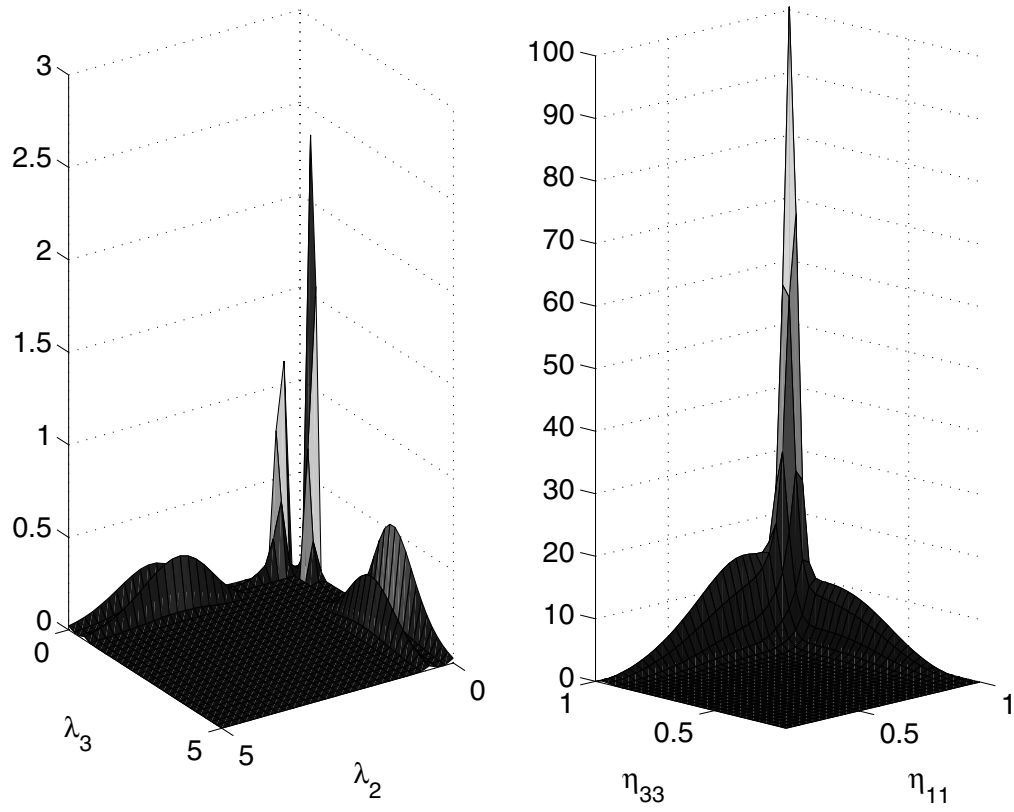


Figure 1. Bivariate Marginals of the unsupervised mixture importance density used in Section 5.1.

ratios the straightforward Gibbs sampler and the random permutation sampler are compared in Table 1 for the 3-state inhomogeneous Poisson process to be studied in Section 5.1. Whereas the Gibbs sampler misses most of the modes, random permutation sampling is nearly balanced.

4. A SIMULATION EXPERIMENT AND COMPARISON TO ALTERNATIVE METHODS

4.1. The simulation experiment

For illustration, we consider a small simulation experiment. Fifty data sets consisting of 100 bivariate observations were simulated from the bivariate mixture model

$$y_t \sim \sum_{k=1}^{K} \eta_k N\left(y_t; \mu_k, \sigma^2 E\right), \tag{22}$$

where $K = 4$, $\eta = (\eta_1, \ldots, \eta_4)' = (0.1, 0.24, 0.26, 0.4)'$, $\mu_1 = (-2, -2)'$, $\mu_2 = (-2, 2)'$, $\mu_3 = (2, -2)'$, $\mu_4 = (2, 2)'$, and E is the identity matrix. $\sigma^2 = 0.01$ is chosen in such a way

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Table 1. Searching for non-visited modes.

			$q\ (\rho_r(\theta^\star, \eta^\star))\ /\ q\ (\theta^\star, \eta^\star)$	
$\rho_r(\cdot)$			Gibbs sampling	Random permutation sampling
3	2	1	0.0000	0.8514
2	3	1	0.3055	0.9034
3	1	2	0.0000	1.0684
1	3	2	0.1255	0.9589
2	1	3	0.0000	0.9186
1	2	3	1.0000	1.0000

Table 2. Performance of various methods of estimating the marginal likelihood log $p(y|K)$ for $K = 4$ over 50 data sets simulated from a mixture of four bivariate normal distributions.

Estimator	Average Bias B	Average MSE
$\log \hat{p}_{IS}(y)$	0.0001	0.00001
$\log \hat{p}_{RI}(y)$	−0.008	0.00007
$\log \hat{p}_{BS}(y)$	−0.004	0.00002
$\log \hat{p}_{CH}(y)$	−3.178	10.1
$\log \hat{p}^{\star}_{CH}(y)$	0.	0.
$\log \hat{p}_{\mathcal{R}_1}(y)$	−0.787	1.24
$\log \hat{p}^{\star}_{\mathcal{R}_1}(y)$	7.69	64.97
$\log \hat{p}_{\mathcal{R}_2}(y)$	−0.0085	0.00008

that the observations from the various groups are totally well-separated. η and $\mu_1, \ldots, \mu_K$ are estimated from the data based on the priors $\mu_j \sim N(0, 4E)$ and $\eta \sim D(4, \ldots, 4)$, whereas σ^2 is assumed to be known when fitting model (22) to the various data sets. This allows *exact* computation of the marginal likelihood $p(y)$ for $K = 4$ for each of the 50 data sets and provides a 'gold standard' against which various methods of *estimating* the marginal likelihood could be compared.

To evaluate a certain estimator $\hat{p}(y)$ of the marginal likelihood $p(y)$ we will use two measures of performance, first the relative bias B which is the average estimation error of the log of the marginal likelihood $\log \hat{p}(y) - \log p(y)$ over the 50 replications of this data set and the relative mean-squared error MSE which is the average of the squared estimation error $(\log \hat{p}(y) - \log p(y))^2$, where log is the natural logarithm.

We computed the bridge sampling estimator $\hat{p}_{BS}(y)$, the importance sampling estimator $\hat{p}_{IS}(y)$ and the reciprocal importance sampling estimator $\hat{p}_{RI}(y)$ based on the importance density (21) which is constructed from the MCMC output of a random permutation sampler with $M = L = 3000$ and $S = 50 \cdot 4!$. The various estimators are compared in Table 2 through bias and relative mean-squared error.

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4.2. Chib's method

Chib's estimator of the marginal likelihood (1), suggested in Chib (1995), is based on the identity (2) and reads:

$$\hat{p}_{CH}(y) = \frac{p(y|\phi^\star)p(\phi^\star)}{\hat{p}(\phi^\star|y)}. \tag{23}$$

$\phi^\star$ is an arbitrary point in a region of high posterior density and could be determined as that MCMC draw $\phi^{(m)}$ which maximizes the non-normalized posterior $p(y|\phi^{(m)})p(\phi^{(m)})$. The posterior ordinate $\hat{p}(\phi^\star|y)$ is also estimated from the MCMC output. For simple models, where it is possible to draw ϕ, conditional on knowing I^N, within one block from a density $p(\phi|I^N, y)$ with known normalizing constant, one obtains:

$$\hat{p}(\phi^\star|y) = M^{-1}\sum_{m=1}^{M} p\big(\phi^\star|(I^N)^{(m)}, y\big), \tag{24}$$

where $(I^N)^{(m)}$, $m = 1, \ldots, M$ are the MCMC draws of the latent variable I^N. For more general cases see Chib (1995) and Chib and Jeliazkov (2001).

Both in Chib (1995) and Chib (1996), this method was applied to estimate the unconstrained marginal likelihood (16) for simple Markov switching models. For illustration we apply the method of Chib (1995) also to the synthetical data discussed in Section 4.1. Despite the simplicity of the model (the whole parameter ϕ may be sampled within one block), the resulting estimator $\hat{p}_{CH}(y)$ shows considerable bias in Table 2 over 50 replications of this data set.

Where does the bias come from? Integration in the definition of the marginal likelihood through (16) is over the unconstrained parameter space $\mathcal{A}$ which is the union of $K!$ subspaces $\mathcal{L}_r$, $r = 1, \ldots, K!$, differing only in the labeling of the K states of I_t. Therefore the unconstrained marginal likelihood may be expressed as:

$$\begin{aligned} p(y) &= \sum_{s=1}^{K!}\int_{\mathcal{L}_s} p(y_1, \ldots, y_N|\psi)p(\psi)\nu(d\psi) \\ &= K!\int_{\mathcal{L}_{s_0}} p(y_1, \ldots, y_N|\psi)p(\psi)\nu(d\psi), \end{aligned} \tag{25}$$

where s_0 is an arbitrary value between 1 and $K!$. Thus to compute the unconstrained marginal likelihood we need simulations from *all* labelling subspaces. Alternatively, if we had simulations from a unique labelling subspace, only, we obtain the unconstrained marginal likelihood by multiplying with the factor $K!$. For all data sets in our simulation study, for instance, the modes of the unconstrained posterior are extremely well-separated and the Gibbs sampler sticks at one of the labelling subspaces $\mathcal{L}_{s_0}$ instead of exploring *all* labelling subspaces. Consequently $\hat{p}_{CH}(y)$ estimates

$$\int_{\mathcal{L}_{s_0}} p(y_1, \ldots, y_N|\psi)p(\psi)\nu(d\psi)$$

instead of $p(y)$, and consequently the expected bias is equal to:

$$E(\log \hat{p}_{CH}(y) - \log p(y)) = -\log K! \tag{26}$$

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As expected from (26), the bias of Chib's estimator $\log \hat{p}_{CH}(y)$ in Table 2 turns out to be equal to $-\log K\,! = -\log 24 = -3.178$. If we correct for this bias,

$$\log \hat{p}^{\star}_{CH}(y) = \log \hat{p}_{CH}(y) + \log\ K! \tag{27}$$

we obtain an estimator which gives the precise answer for all data sets (see Table 2). The problem with the bias corrected estimator $\hat{p}^{\star}_{CH}(y)$, however, is that in cases where *occasional* label switching occurs as in the case study to be discussed in Section 5.1, it is no longer possible to quantify the bias and to correct for it.

4.3. Estimating constrained marginal likelihoods

Applied econometricians often prefer to estimate marginal likelihoods from constrained models (see, e.g. Kaufmann 2000; Lenk and DeSarbo 2000; Sögner 2001). The constraint is included to achieve formal identification of the model which is identified only up to permutations of the labels of the states I_t (see Celeux *et al*. 2000; Stephens 2000; Frühwirth-Schnatter 2001b).

In this case the marginal likelihood is defined as the normalizing constant of the posterior of the constrained model:

$$p_{\mathcal{R}}(y) = \int_{\mathcal{R}} p(y_1, \ldots, y_N|\psi) p_{\mathcal{R}}(\psi) \nu(d\psi), \tag{28}$$

where $\mathcal{R}$ is the subspace on which the constraint holds. Theoretically the inclusion of a constraint will not change the marginal likelihood, if we define the constrained prior $p_{\mathcal{R}}(\psi)$ from an unconstrained prior $p(\psi)$ which is invariant to permuting the labels of the states by

$$p_{\mathcal{R}}(\psi) \propto p(\psi) I_{\mathcal{R}}(\psi),$$

where $I_{\mathcal{R}}(\psi)$ is 1 iff $\psi \in \mathcal{R}$. Then

$$p_{\mathcal{R}}(\psi) = K!\, p(\psi)$$

and we obtain that both approaches lead to the same marginal likelihood:

$$p_{\mathcal{R}}(y) = p(y).$$

Therefore for model selection through marginal likelihoods the inclusion of an identification constraint $\mathcal{R}$ is not necessary. We now discuss in detail why we prefer the unconstrained marginal likelihood (16) to the one defined from the constrained marginal likelihood (28). This is a computational rather than a theoretical issue since theoretically the two marginal likelihoods will be the same.

We will give details for the synthetical data generated in Section 4.1. We start with estimating the constrained marginal likelihood (28) under the following constraint on the weights:

$$\mathcal{R}_1 : \eta_1 < \cdots < \eta_4, \tag{29}$$

which is applied in Aitkin and Rubin (1985) and Lenk and DeSarbo (2000) as a standard constraint for this type of mixture models. For estimation we apply the reciprocal importance sampling estimator (7) as Lenk and DeSarbo (2000). To obtain the constrained marginal likelihood the prior has to be normalized over the constrained space, with the normalizing constant being equal

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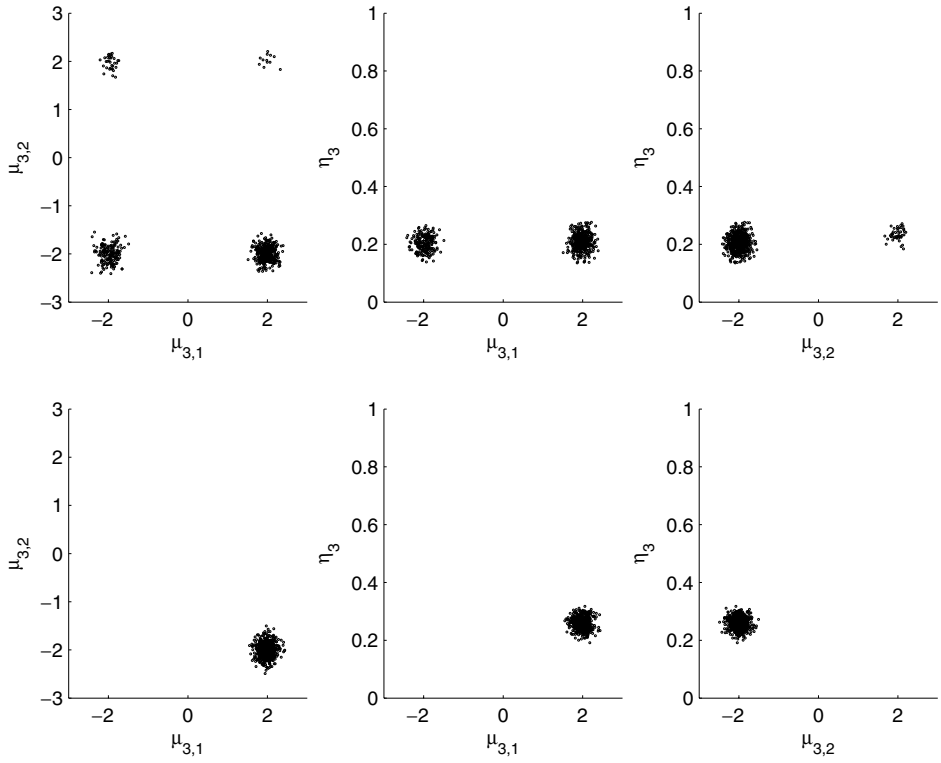


Figure 2. Simulations from the various bivariate marginal densities $p(\mu_{3,1}, \mu_{3,2}|y)$, $p(\mu_{3,1}, \eta_3|y)$ and $p(\mu_{3,2}, \eta_3|y)$ for $K = 4$; first row: permutation sampling under constraint (29), second row: slice sampling under constraint (29).

to $1/K!$:

$$\hat{p}_{\mathcal{R}}(y) = K! \left\{ M^{-1} \sum_{m=1}^{M} \frac{q\left(\phi^{(m)}\right)}{p^{\star}\left(\phi^{(m)}|y\right)} \right\}^{-1}, \tag{30}$$

where $\phi = (\mu_1, \ldots, \mu_K, \eta_1, \ldots, \eta_K)$. $q(\phi)$ is a unimodal importance density fitted to the MCMC draws $\phi^{(m)}$, $m = 1, \ldots, M$, by matching moments between density and draws. In Lenk and DeSarbo (2000), the estimator (30) is based on draws of a slice sampler under constraint (29). The resulting estimator is denoted by $\hat{p}_{\mathcal{R}_1}(y)$ in Table 2 and exhibits considerable bias which results from a lack of appropriate mixing of the slice sampler. In Figure 2, draws from the slice sampler under constraint (29) are compared to draws from the permutation sampler under the same constraint, which has been shown in Frühwirth-Schnatter (2001b) to be well-mixing with respect to the constrained posterior $p_{\mathcal{R}_1}(\phi|y)$. From the first row of this figure we see that the constrained posterior has multiple, non-symmetric modes. Apparently the slice sampler mixes badly and remains within a single modal region, only. This introduces a bias when estimating the marginal likelihood from the draws of the slice sampler.

It does not help to base the estimator (30) on the output of a sampler that is well-mixing under constraint (29). As the constrained posterior has more than one mode whereas the importance

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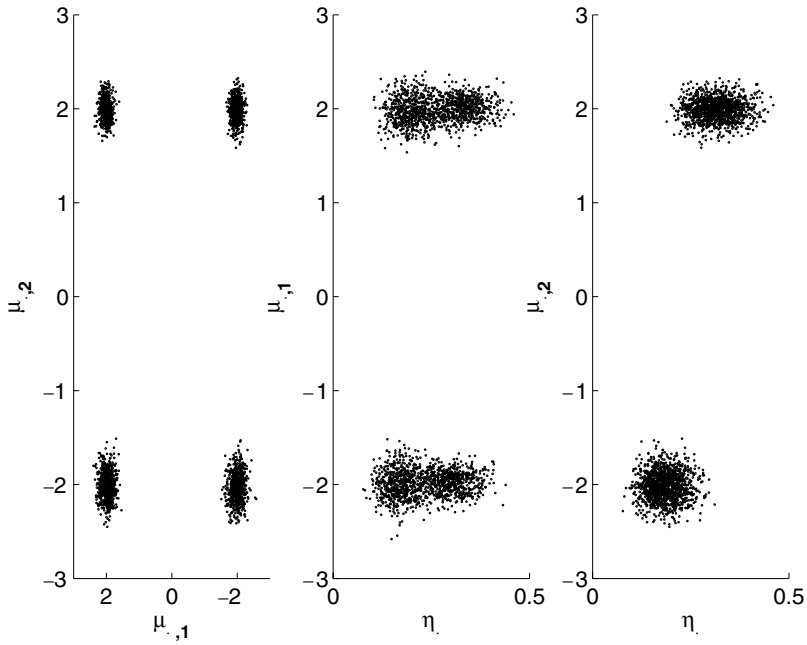


Figure 3. Simulations from the bivariate marginal density $p(\mu_{\cdot,1}, \mu_{\cdot,2}|y)$, $p(\mu_{\cdot,1}, \eta_{\cdot}|y)$ and $p(\mu_{\cdot,2}, \eta_{\cdot}|y)$ of the unconstrained posterior for $K = 4$ obtained by random permutation sampling.

density fitted to the draws is unimodal, the resulting estimator which is denoted by $\hat{p}^{\star}_{\mathcal{R}_1}(y)$ in Table 2 exhibits extreme standard errors. Bias and high standard errors, however, disappear, if the estimator (30) is based on a constraint that respects the geometry of the posterior distribution. For the example at hand, such a constraint may be found by exploring the draws from the random permutation sampler in Figure 3. Each cluster in this figure corresponds to one group. From the marginal distributions in Figure 3 involving the weights η it becomes apparent that the clusters are not well-separated in direction of η and constraint (29) fails to identify the four groups. The clusters, however, are well-separated in the direction of the means and one possible way of identifying four groups is to separate the four clusters by following constraint on the means:

$$\mathcal{R}_2 : \max(\mu_{1,1}, \mu_{2,1}) < \min(\mu_{3,1}, \mu_{4,1}), \quad \mu_{1,2} < \mu_{2,2}, \mu_{3,2} < \mu_{4,2}. \tag{31}$$

As can be seen from Table 2 the resulting estimator $\hat{p}_{\mathcal{R}_2}(y)$ has practically no bias and is extremely precise. These empirical results indicate that the reliability of estimating a constrained marginal likelihood depends very much on selecting a sensible constraint.

5. EMPIRICAL STUDIES

5.1. Testing for the inhomogeneity of a discrete time Poisson process

We reanalyze an example given in Chib (1996) which is based on data analyzed in Leroux and Puterman (1992). The data are the number y_t of movements by a fetal lamb in 240 consecutive

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Table 3. Various estimates of the log of marginal likelihood $\log p(y|K)$ for $K = 2$ and $K = 3$ (relative standard error are given in parenthesis).

Estimator	$K = 2$	$K = 3$
$\log \hat{p}_{BS}(y\|K)$	−185.077 (0.0037)	−179.138 (0.0018)
$\log \hat{p}_{IS}(y\|K)$	−185.075 (0.0028)	−179.108 (0.0027)
$\log \hat{p}_{RI}(y\|K)$	−185.082 (0.0045)	−179.989 (0.0128)
$\log \hat{p}_{CF}(y\|K)$	−185.741 (0.0128)	−181.374 (0.0881)

five-second intervals. y_t is modelled as an inhomogeneous Poisson process where the intensity can change according to a K-state Markov switching variable I_t:

$$y_t|\lambda_1, \ldots, \lambda_K, I^N \sim \text{Poi}(\lambda_{I_t}).$$

In the present paper we will discuss the issue of formal testing for inhomogeneity and choosing the number K of states by comparing the corresponding models through the unconstrained marginal likelihood $p(y|K)$. Estimation is based on the following symmetric priors: $\lambda_j \sim G(1, \beta)$ with $\beta = 0.5$ and $\eta_{i\cdot} \sim D(e_{0,i1}, \ldots, e_{0,iK})$ with $e_{0,ii} = \alpha$ where $\alpha = 2$ and $e_{0,ij} = 1/(K-1)$, if $i \neq j$. For $K = 1$ the marginal likelihood is known analytically:

$$p(y|K=1) = \frac{b^a \Gamma\left(a + \sum_{t=1}^N y_t\right)}{(b+N)^{(a+\sum_{t=1}^N y_t)} \Gamma(a) \prod_{t=1}^N \Gamma(y_t+1)},$$

leading to $\log p(y|K=1) = -204.25$.

For $K > 1$ we use the various estimators of $p(y|K)$ discussed above. We computed the bridge sampling estimator $\hat{p}_{BS}(y)$, the importance sampling estimator $\hat{p}_{IS}(y)$ and the reciprocal importance sampling estimator $\hat{p}_{RI}(y)$ based on the importance density (21) which is constructed from the MCMC output of a random permutation sampler with $M = L = 5000$ and $S = 100 \cdot K!$. The resulting marginal likelihoods are summarized in Table 3. Importance sampling and bridge sampling lead to nearly identical estimates for $\log p(y|K)$ with comparable standard errors, whereas reciprocal importance sampling exhibits higher standard errors than these two estimators.

Concerning the number of states, there is overwhelming evidence against the hypothesis that the process is homogeneous. There is clear evidence in favour of the three-state Markov switching model compared to the two-state Markov switching model. This evidence is insensitive to the underlying prior as can be seen from Table 4 where the log Bayes factors $\log \hat{L}_{BS}(y^N|K=1) - \log \hat{L}_{BS}(y^N|K=2)$ and $\log \hat{L}_{BS}(y^N|K=2) - \log \hat{L}_{BS}(y^N|K=3)$ are reported for different assumptions concerning the priors.

For illustration, we also computed the estimator of Chib (1995) given by (23), where $\phi = (\lambda_1, \ldots, \lambda_K, \eta)$. From Table 3 we see again that this estimator is biased in comparison to the other methods. As for $K = 2$ no label switching occurred, it is possible to correct for this bias as in Section 4.2:

$$\log \hat{p}^{\star}_{CH}(y|K=2) = \log \hat{p}_{CH}(y|K=2) + \log(2) = -185.048.$$

Bias correction is not possible for $K = 3$. It is evident from Figure 4 that for $K = 3$ the unconstrained Gibbs sampler shows frequent label switching not only in the MCMC draws of $\lambda_1^{(m)}, \ldots, \lambda_K^{(m)}$, but also in the functional values of the conditional densities $p(\phi^{\star}|(I^N)^{(m)}, y)$ in

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Table 4. Sensitivity analysis with respect to the prior $\eta_{i\cdot} \sim D(e_{0,i1}, \ldots, e_{0,iK})$ with $e_{0,ii} = \alpha$ and $e_{0,ij} = 1/(K-1)$, if $i \neq j$, and $\lambda_j \sim G(1, \beta)$.

	$\log \hat{p}_{BS}(y\|K=1) - \log \hat{p}_{BS}(y\|K=2)$				$\log \hat{p}_{BS}(y\|K=2) - \log \hat{p}_{BS}(y\|K=3)$			
α	$\beta = 0.1$	$\beta = 0.5$	$\beta = 1$	$\beta = 2$	$\beta = 0.1$	$\beta = 0.5$	$\beta = 1$	$\beta = 2$
1	−17.6047	−18.2787	−17.9367	−16.9325	−4.1212	−5.3158	−5.4567	−5.0853
2	−18.5326	−19.1739	−18.8582	−17.8861	−4.7600	−5.9399	−6.0876	−5.6488
5	−19.2147	−19.8739	−19.5495	−18.6505	−5.5966	−6.7664	−6.8728	−6.2955
10	−19.2650	−19.9059	−19.5810	−18.7404	−6.1760	−7.3447	−7.4233	−6.7240

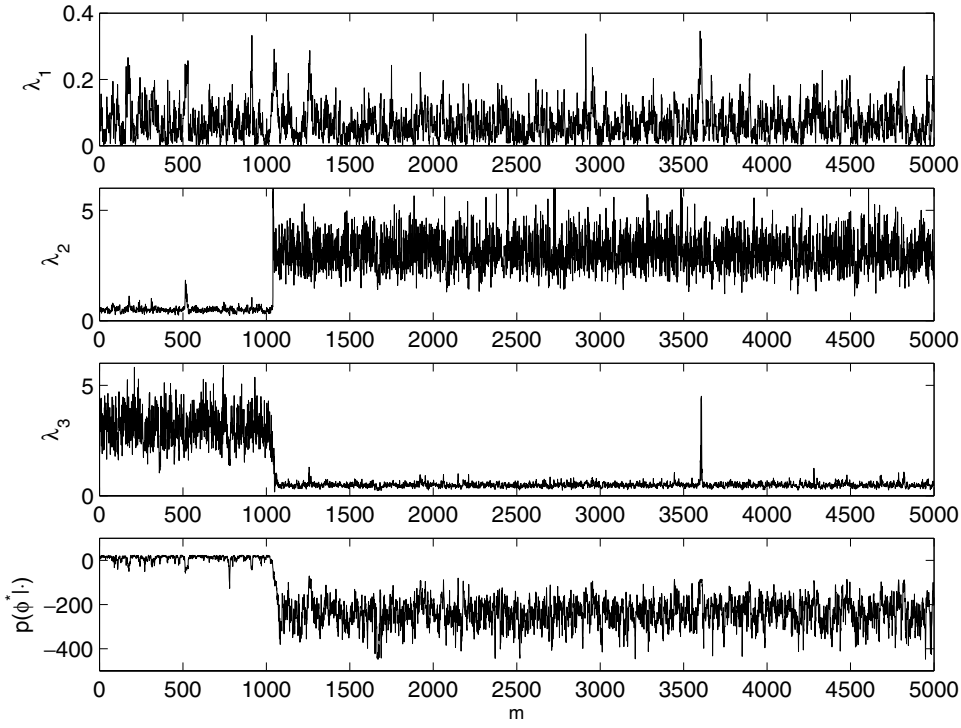


Figure 4. Label switching in the MCMC output $\lambda_1^{(m)}, \ldots, \lambda_K^{(m)}$ and in the functional value $p(\phi^\star|(I^N)^{(m)}, y)$ of the conditional posterior.

(24). This causes an uncontrollable bias when estimating the marginal likelihood in (24) as the average of the functional value $p(\phi^\star|(I^N)^{(m)}, y)$.

5.2. Testing for the presence of a hidden Markov switching process and model order selection in the MSAR model

5.2.1. The data and the models to be compared. We reanalyze a standard time series which has appeared in numerous papers on Markov switching models since the influential paper of

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Hamilton (1989), namely the percentage growth of the U.S. quarterly real GNP series for the period 1951.II to 1984.IV: $y_t = 100\,(\log(\mathrm{GNP}_t) - \log(\mathrm{GNP}_{t-1}))$. The aim of the analysis is to separate those two states of the economy which correspond to recession (negative growth rate) and boom (positive growth rate). Previous studies have produced somewhat conflicting evidence concerning the presence or absence of Markov-switching behaviour in this time series (see Kim and Nelson 1998). We will reconsider this topic based on three different models. The first model is a classical AR(p)-model of unknown model order p without switching:

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + c + \varepsilon_t, \varepsilon_t \sim \mathrm{N}(0, \sigma^2). \tag{32}$$

The second model is the K-state Markov switching autoregressive model, MSAR model for short, with switching mean, switching AR-parameters and switching error variance ('totally switching') applied by McCulloch and Tsay (1994), which takes the form:

$$y_t = \begin{cases} \phi_1^{[1]} y_{t-1} + \cdots + \phi_p^{[1]} y_{t-p} + c^{[1]} + \varepsilon_t, & \varepsilon_t \sim \mathrm{N}\big(0, \sigma_1^2\big), \quad I_t = 1, \\ \vdots & \\ \phi_1^{[K]} y_{t-1} + \cdots + \phi_p^{[K]} y_{t-p} + c^{[K]} + \varepsilon_t, & \varepsilon_t \sim \mathrm{N}\big(0, \sigma_K^2\big), \quad I_t = K. \end{cases} \tag{33}$$

The third model is the K-state MSAR model with switching mean but state independent AR-parameters and state independent variance applied by Chib (1996), which takes the form:

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \begin{cases} c^{[1]} + \varepsilon_t, & \varepsilon_t \sim \mathrm{N}(0, \sigma^2), \quad I_t = 1, \\ \vdots & \\ c^{[K]} + \varepsilon_t, & \varepsilon_t \sim \mathrm{N}(0, \sigma^2), \quad I_t = K. \end{cases} \tag{34}$$

For all model structures we assume that the model order p varies between 0 and 4 and that K is equal to 2 or 3, leading to a total of twenty different Markov switching models.

The priors are selected to be rather vague and state independent. We assume no prior correlation among the regression parameters. The prior on the switching intercept is $N(0, 1)$, the prior both on switching and state independent AR-parameters is $N(0, 0.25)$. The prior both on switching and state independent variances is $IG(2, 0.5)$. The prior on the transition probability distribution η_i is $D(e_{0,i1}, \ldots, e_{0,iK})$ with $e_{0,ii} = 2$ and $e_{0,ij} = 1/(K-1)$, if $i \neq j$.

5.2.2. Model comparison. The marginal likelihoods are estimated from the MCMC output of a random permutation sampler ($M = 6000$ after a burn-in phase of 1000 simulations) using the bridge sampling estimator where the construction of the importance density $q(\phi)$ according to (21) is based on a rather high number of components ($S = 100 \cdot K!$). It should be once more emphasized that the construction of the importance density is unsupervised and that the importance densities differ between the different models.

From Table 5 reporting the log marginal likelihoods we find that the 'best' model in terms of having the highest marginal likelihood among all models is the two-state totally switching MSAR-Model of order $p = 2$. This result is interesting for various reasons: first, we were able to produce evidence in favour of Markov switching heterogeneity from univariate time series observations of the GDP alone, without the need to include other time series such as in Kim and Nelson (1998). Second, the evidence in favour of the hypothesis that the dynamic pattern of the economy is different between contraction and expansion periods confirms the empirical results of McCulloch and Tsay (1994).

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Table 5. Log of marginal likelihoods log $p(y|K, p)$ for various Markov switching models of different order p and different number of states K, estimated by bridge sampling.

		Switching mean		Totally switching	
p	$K = 1$	$K = 2$	$K = 3$	$K = 2$	$K = 3$
0	−199.71	−193.54	−192.25	−194.25	−193.10
1	**−194.22**	**−192.54**	−192.75	−193.58	−194.71
2	−196.30	−194.15	−194.38	**−191.62**	−194.33
3	−197.26	−194.59	−194.74	−193.67	−196.78
4	−199.18	−195.70	−195.72	−195.34	−199.88
Random	−195.66	−193.60	−193.25	−192.918	−194.29

Most importantly, we find from Table 5 that testing for Markov switching heterogeneity is highly influenced by the selecting the appropriate model order. Among all classical AR models without switching the optimal order is $p = 1$. If we compared this non-switching model of optimal order with a two-state totally switching model of order four, which is the model considered by McCulloch and Tsay (1994), we would end up with evidence in favour of *no* Markov switching heterogeneity. From Table 5 we see that for a two-state totally switching MSAR-model the optimal model order is $p = 2$ rather than $p = 4$. Only if we compared the non-switching model of optimal order with a two-state switching model with p close to the optimal order, we end up with evidence *in favour of* Markov switching heterogeneity. This result indicates the importance of simultaneously testing for Markov switching heterogeneity and selecting the appropriate model order and might explain why previous papers produced conflicting evidence from this GDP time series.

In a similar way the model order influences the decision between a model with switching mean and a totally switching model. To reduce the ambiguity due the model order, we include the order p as a random quantity into each model class and assume that p is uniformly distributed on $\{0, 1, \ldots, 4\}$ *a priori*. The model likelihood for each model class is then obtained from the marginal likelihoods where p is assumed to be known:

$$p(y|K) = \frac{1}{5}\sum_{p=0}^{4} p(y|K, p).$$

The log of this marginal likelihood is reported in the last row of Table 5. It shows clear evidence in favour of all Markov switching models when compared to the non-switching autoregressive model. The two-state totally switching model remains to be the best Markov switching model among the tested ones, even if we take the uncertainty in quantifying the appropriate model order into account.

5.2.3. Comparing the various estimators. We finish by comparing the bridge sampling estimator $\log \hat{p}_{BS}(y|\mathcal{M})$, the importance sampling estimator $\log \hat{p}_{IS}(y|\mathcal{M})$ and the reciprocal importance sampling estimator $\log \hat{p}_{RI}(y|\mathcal{M})$ discussed in Section 2 for two different importance densities, being equal to the mixture density (21) with $S = M_0 K!$. We compare a 'fine' importance density based on a rather high number of components ($M_0 = 100$) with a 'course' importance density based on a low number of components ($M_0 = 10$). Table 6 summarizes the estimated

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Table 6. Log of estimated marginal likelihoods log $\hat{p}(y)$ for various Markov switching models differing in K and p; the estimates $\hat{p}_{BS}(y)$ (BS), $\hat{p}_{IS}(y)$ (IS) and $\hat{p}_{RI}(y)$ (RI) are based on importance density (21) with $S = M_0 K!$.

		Switching mean				Totally switching			
		$K = 2, M_0 =$		$K = 3, M_0 =$		$K = 2, M_0 =$		$K = 3, M_0 =$	
p		100	10	100	10	100	10	100	10
0	BS	−193.54	−193.54	−192.25	−192.1	−194.25	−194.24	−193.10	−192.85
	IS	−193.53	−193.54	−191.91	−191.98	−194.22	−194.21	−192.92	−192.75
	RI	−193.56	−193.55	−194.76	−193.87	−194.26	−194.26	−196.57	−195.5
1	BS	−192.54	−192.53	−192.75	−192.39	−193.58	−193.57	−194.71	−193.92
	IS	−192.52	−192.47	−192.29	−192.19	−193.58	−193.5	−194.08	−193.7
	RI	−192.56	−192.53	−197.56	−194.1	−193.94	−193.94	−200.14	−198.92
2	BS	−194.15	−194.12	−194.38	−194.1	−191.62	−191.64	−194.33	−193.1
	IS	−194.15	−193.19	−194.02	−193.6	−191.61	−191.62	−192.87	−193.32
	RI	−194.23	−194.17	−197.32	−196.02	−192.07	−193.32	−200.26	−198.18
3	BS	−194.59	−194.58	−194.74	−194.49	−193.67	−193.6	−196.78	−195.19
	IS	−194.57	−194.59	−194.68	−194.56	−193.61	−193.64	−194.43	−194.96
	RI	−194.61	−194.62	−198.26	−197.16	−195.98	−196.04	−205.36	−202.64
4	BS	−195.7	−195.69	−195.72	−195.4	−195.34	−195.27	−199.88	−197.33
	IS	−195.65	−195.66	−195.6	−195.44	−195.26	−195.17	−198.04	−198.31
	RI	−195.76	−195.79	−199.93	−198.06	−198.5	−196.3	−208.67	−206.2

marginal likelihood, whereas Table 7 summarizes the relative standard errors (in percent) of the various estimators for the two different importance densities for all models.

As expected from the discussion at the end of Section 2.2, the bridge sampling estimator turns out to be more robust with respect to the different importance densities than the other methods. In terms of standard errors, the bridge sampling estimator outperforms the other estimators for each of the 20 models both for $M_0 = 10$ and $M_0 = 100$. In some cases (e.g. switching mean, $K = 2$, $p = 2$, $M_0 = 100$) importance sampling nearly gives the same standard error as bridge sampling, in other cases (e.g. switching mean, $K = 2$, $p = 2$, $M_0 = 10$) the relative standard error of importance sampling is 79.3% compared to 1.7% for the bridge sampling estimator. Similarly, in some cases (e.g. switching mean, $K = 2$, $p = 3$, $M_0 = 10$) reciprocal importance sampling gives nearly the same standard error as bridge sampling, in other cases (e.g. totally switching, $K = 2$, $p = 3$, $M_0 = 10$) the relative standard error of reciprocal importance sampling is 111.8% compared to 2.7% for the bridge-sampling estimator. For all estimators standard errors are higher for over-identified models with $K = 3$ than for more parsimonious models with $K = 2$.

6. CONCLUDING REMARKS

In the present paper we discussed Bayesian model selection for mixture and Markov switching models, based on the marginal likelihood derived from the unconstrained posterior. We suggested computing the marginal likelihood by the technique of bridge sampling.

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Table 7. Comparing bridge sampling, importance sampling and reciprocal importance sampling estimators by their relative standard errors (in percent); the estimates $\hat{p}_{BS}(y)$ (BS), $\hat{p}_{IS}(y)$ (IS) and $\hat{p}_{RI}(y)$ (RI) are based on importance density (21) with $S = M_0 K!$.

		Switching mean				Totally switching			
		$K = 2, M_0 =$		$K = 3, M_0 =$		$K = 2, M_0 =$		$K = 3, M_0 =$	
p		100	10	100	10	100	10	100	10
0	BS	0.43	0.48	1.6	4.3	0.43	0.87	3.4	2.1
	IS	0.58	0.61	17.9	6.8	1.1	2.0	13.2	9.8
	RI	0.86	1.1	25.1	50.7	0.67	1.9	39.0	27.8
1	BS	1.1	1.1	2.8	2.8	2.6	1.7	5.2	5.4
	IS	2.6	5.1	17.1	12.3	3.3	5.8	14.9	56.2
	RI	0.98	2.1	60.5	21.8	13.1	17.0	24.4	39.3
2	BS	1.1	1.7	2.6	3.1	1.5	1.6	5.1	3.8
	IS	1.7	79.3	14.0	40.4	2.7	2.7	29.6	24.3
	RI	3.9	2.6	29.8	18.1	19.0	100.4	36.6	32.4
3	BS	1.6	2.1	2.7	2.8	2.2	2.7	6.0	6.8
	IS	2.8	3.0	9.2	19.1	4.3	3.8	72.4	40.2
	RI	2.3	2.6	26.9	56.7	89.4	111.8	60.5	45.7
4	BS	1.6	2.0	2.5	3.6	2.6	2.4	8.4	5.3
	IS	2.6	3.8	6.1	8.1	8.6	15.2	34.2	20.5
	RI	2.5	4.2	49.7	36.5	105.3	11.5	74.1	49.9

We limited the discussion to those models, where an explicit formula for the likelihood $p(y_1, \ldots, y_N|\theta, \eta)$ with the hidden indicator process I^N being integrated out, is available. The method, however, may also be extended to more general models, for instance to the important case, where additionally to the discrete latent process I^N a continuous latent state process x^N is present. This has been actually carried out by Frühwirth-Schnatter (2001a) for the case of regime switching state space models. A further limitation was to the 'Gibbsian' case were the densities appearing in the transition kernel $K(\psi|\psi')$ (19) are standard densities with explicit normalizing constants, only. The method, however, may be extended to MCMC sampler including Metropolis-Hastings-steps, as demonstrated in Kaufmann and Frühwirth-Schnatter (2002) for selecting the number of states and model order selection for switching ARCH models.

A further remark concerns the prior $p(\theta)$ of the state specific parameter θ. In our case studies we used for state specific parameters θ_i priors of the type $\theta_i \sim p(\theta_i|\xi)$ where ξ was a fixed hyperparamter. An alternative method which has been shown to reduce sensitivity of model selection to specific choices of ξ is to treat ξ as an unknown hyperparameter and to add an additional prior on ξ as in Richardson and Green (1997) and Robert *et al.* (2000). The estimators of the marginal likelihood suggested in Section 2 are easily extended to this more general setting, as both the likelihood $p(y_1, \ldots, y_N|I^N, \theta)$ as well as the prior $p(I^N|\eta)$ are independent of ξ given θ and η. No modification is necessary, if an analytical expression for the marginal prior,

$$p(\theta) = \int \prod_{i=1}^{K} p(\theta_i|\xi) p(\xi)$$

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appearing in the estimators (6)–(8) is available. Otherwise the hyperparameter ξ could be added to the set of unknown parameters before applying bridge sampling technique: $\phi = (\theta, \eta, \xi)$. As ξ is independent of I^N given θ, the importance density $q(\theta, \eta, \xi) = q(\xi)q(\theta, \eta)$, where $q(\theta, \eta)$ is equal to the mixture importance density (21) and $q(\xi)$ is obtained by fitting a unimodal importance density to the MCMC draws $\xi^{(m)}$ will be a sensible choice.

Finally, bridge sampling as a robust method of estimating integrals may be useful not only to approximate marginal likelihood for mixture and Markov switching models, but also in more general fields of econometric inference. Recently, Koopman and Shephard (2002) developed diagnostic methods to test the assumptions behind importance sampling and found empirical evidence for the non-existence of the relevant variance when fitting a stochastic volatility model without leverage to returns of daily Standard & Poor's 100 stock indices. Here bridge sampling is a promising alternative to importance sampling.

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